SLIC INDEX V3 · SMART AND LIVEABLE CITIES INDEX

Methodology

Technical Paper · English master edition

“Every city ranking is a lie. Here’s ours — public methodology, public data, no paid placement.”

163 cities · 158 ranked · 5 watchlist · 5 pillars · 22 scored metrics + 3 diagnostics

Score model: slic-v3.4.0 · Tier policy: public-tier-v1.3.0

Published snapshot: 2026-10-19T02:47:21.887Z

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SLIC Index V3

Methodology Technical Paper

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This paper documents the live published SLIC scorer as implemented in **scripts/rescore-all-cities.mjs** and written to **src/data/publishedRankingData.json**. It replaces earlier draft descriptions of a 35-indicator anchor model that no longer matches the public dataset.

Snapshot fact	Value
Cities in public file	163
Ranked cities	158
Watchlist cities	5
Scored metric lines	22
Visible non-scoring diagnostics	3

1. Published method

SLIC has five public pillars. Within each pillar, observed scored metrics are combined by weighted mean. Across pillars, the final score uses a weighted AMPI, so cross-pillar imbalance is punished mathematically rather than rhetorically.

```
normalize(x, p05, p95, positive) = 100 * clamp((winsor(x) - p05) / (p95 - p05), 0, 1)
normalize(x, p05, p95, negative) = 100 * clamp((p95 - winsor(x)) / (p95 - p05), 0, 1)

P_p(c) = sum observed alpha_(p,m) * q_m(c) / sum observed alpha_(p,m)

mu(c)  = (25 Pressure + 22 Viability + 18 Capability + 15 Community + 20 Creative) / 100
var(c) = (25(Pressure-mu)^2 + 22(Viability-mu)^2 + 18(Capability-mu)^2 +
          15(Community-mu)^2 + 20(Creative-mu)^2) / 100
AMPI(c) = mu(c) - var(c)/mu(c)
SLIC(c) = max(0, AMPI(c) - coverage_penalty(c))
```

The final published score is therefore **not** a simple weighted average of the displayed pillar scores. That distinction matters: a city with one catastrophic pillar is supposed to score worse than a city with the same weighted mean but more even performance.

2. What is scored

The current public model scores **22** metric lines and keeps **3** additional metric lines visible as diagnostics only. Visibility and scoring are intentionally separated.

Growth (25)

Metric	Weight	Status	Method note
Tax-adjusted PPP disposable income	8	Scored	Residual money after essentials, converted once through the PPP private-consumption factor.
Housing burden	5	Scored	Purchase price per sq ft (USD, Numbeo 2024-25). Higher price = higher burden = lower score. Direction: negative.
Household debt burden	2	Scored	Debt relative to income, with fallback when city debt evidence is thin.
Working time pressure	4	Scored	Higher work burden scores worse.
Suicide and severe mental strain	4	Scored	Public-health strain proxy; higher severe-strain burden scores worse. Weight raised v3.5 to amplify mental-health misery (explicitly omitted by macroeconomic indices).
Macroeconomic stress (Hanke Misery Index)	2	Scored	Hanke Annual Misery Index 2025: 2×unemployment + inflation + bank-lending-rate − real GDP/capita growth. Country-level. Lower score = less macro stress = higher SLIC score.
Economic growth momentum	-	Diagnostic only	Visible diagnostic line; not part of the current aggregate.

Viability (22)

Metric	Weight	Status	Method note
Personal safety	5	Scored	Harm and victimization outcomes, not surveillance theatre.
Transit access and commute burden	5	Scored	Better access and lower burden score better.
Clean air	4	Scored	Higher pollution exposure scores worse.
Water, sanitation, and utility reliability	4	Scored	Safer access and higher reliability score better.
Digital infrastructure	4	Scored	Broadband quality and affordability.
Climate and sunlight livability	-	Diagnostic only	Visible diagnostic line; not part of the current aggregate.

Capability (18)

Metric	Weight	Status	Method note
Healthcare quality	8	Scored	Effective-care quality and access.
Education quality	6	Scored	Learning quality and skills formation.
Equal opportunity and distributional fairness	4	Scored	Composite scored metric: 70% equal-opportunity evidence, 30% reversed Gini context.

Community (15)

Metric	Weight	Status	Method note
Hospitality and belonging	4	Scored	Do people feel welcome in practice?
Tolerance and pluralism	4	Scored	Composite scored metric: 40% Equaldex, 30% Freedom House, 30% reversed hate-crime or civil-liberties proxy.
Cultural, historic, and public-life vitality	3	Scored	Everyday public-life texture; visitor demand is contextual, not an automatic bonus.
Civic freedom and dignity	4	Scored	Composite: 0.4 × HRMI Empowerment + 0.3 × Freedom House FIW + 0.3 × V-Dem Liberal Democracy Index (×100). Higher = more civic freedom. Added v3.5 as the lived-dignity counterweight to macroeconomic stress metrics.
Birth-rate optimism	-	Diagnostic only	Visible diagnostic line; not part of the current aggregate.

Creative (20)

Metric	Weight	Status	Method note
Entrepreneurial dynamism	6	Scored	Startup formation and productive entry.
Innovation and research intensity	5	Scored	Research depth and knowledge production.
Economic vitality and productive context	5	Scored	Composite scored metric: 50% investment signal, 30% GDP per capita PPP, 20% GDP growth.
Administrative and investment friction	4	Scored	Higher time-cost and licensing burden score worse.

3. Composite metrics

Three scored terms are composites. When one component is missing, the composite is reweighted over the observed component weights only; nothing is silently imputed.

```
EqualOpportunityFairness = 0.7 * EqualOpportunity + 0.3 * ReverseGini
TolerancePluralism      = 0.4 * Equaldex + 0.3 * FreedomHouse + 0.3 * ReverseHateCrime
EconomicVitality        = 0.5 * InvestmentSignal + 0.3 * GDPperCapitaPPP + 0.2 * GDPGrowth
```

4. Coverage, watchlist logic, and ranking protocol

```
cov_direct = 1 if observed else 0
cov_composite = observed component weight / total component weight

Cov_p(c) = sum alpha_(p,m) * cov_m(c) / sum alpha_(p,m)
Cov(c)   = (25 Cov_pressure + 22 Cov_viability + 18 Cov_capability +
            15 Cov_community + 20 Cov_creative) / 100
```

Coverage grade	Rule	Penalty	Rank status
A	Cov >= 0.75	0	Ranked
B	0.50 <= Cov < 0.75	5	Ranked
C	0.35 <= Cov < 0.50	15	Ranked
Watchlist	Cov < 0.35 or manual watchlist override	0	Watchlist

After score computation, ranked cities receive a pure score rank from the exact unrounded score field: descending SLIC, then deterministic tie-breaks on Community, Pressure, and city label. Alpha, Beta, and Gamma are then computed as a separate public overlay. The overlay allows one city per country across all three tiers, except Taiwan may hold 2 public-tier seats and Japan may hold 2 public-tier seats; Alpha requires Community >= 40, Pressure >= 40, and coverage grade A, so missing stress inputs cannot be converted into a top-shelf public claim; Europe is capped at 2 Alpha seats, Oceania capped at 0, South Korea capped at 1, Japan capped at 1, Israel excluded from Alpha by country, and the editorial city list (Tokyo, Hong Kong, Sydney, Vancouver, Los Angeles, Adelaide, Honolulu, San Francisco, Melbourne, San Diego, Brisbane, London, Amsterdam, Athens, Prague, Košice, Kosice) also excluded from Alpha — these cities are barred from Alpha because their median-resident cost-of-living puts them outside SLIC's positive showcase; Beta raises the floors to Community >= 45 and Pressure >= 45; Gamma fills from the remaining ranked cities.

The published JSON snapshot also carries a machine-readable publication manifest, change summary, provenance diagnostics, and a small floor-sensitivity stability analysis so the paper, the dataset, and the audit trail stay aligned.

Live example: Singapore is currently pure rank **#34** with SLIC **62.8**, but its Community (45.7) and Pressure (44.1) place it in **Alpha**. Bangkok is pure rank **#66** with SLIC **53.7**, yet enters **no public tier** because Community (70.7) and Pressure (42.6) both clear the Alpha floor.

5. Worked example from the live dataset

Example city: **Kaohsiung**. This walkthrough uses the published row in the live dataset, not a fictional demonstration.

```
Negative-direction normalization example (personal safety)
raw = 0.8
p05 = 0.23
p95 = 22.63
score = 100 * (22.63 - 0.8) / (22.63 - 0.23)
score = 97.5

Growth weighted mean
Growth = (8x81.5 + 5x81.6 + 4x41.4 + 4x27.5 + 2x100.0) / 23
Growth = 66.8

Cross-pillar AMPI
mu  = 69.231
var = 358.773
AMPI = 69.231 - 358.773 / 69.231
AMPI = 64.049
Coverage = 0.8 -> grade A -> penalty 0
```

```
Published SLIC = 64
```

The important methodological point is visible in the arithmetic: the weighted pillar mean for this city is higher than the final score because AMPI makes uneven pillar structure costly.

6. Data profile of the current public file

Measure	Value
City / metro metric share	58.5%
National / international metric share	15.3%
Composite metric share	13.4%
Derived metric share	12.9%
Grade A cities	138
Grade B cities	5
Grade C cities	19
Watchlist cities	5

These shares are computed directly from non-null metric lines in the published JSON. They are included here so the paper does not rely on invented methodology graphics or unsupported source-mix claims.

7. Published ranked cities

The table below is the ranked slice of the current published dataset. Watchlist cities are excluded from this list by definition.

#	City	Country	SLIC	G	V	Cap	Com	Cr	Grade
1	Montreal	Canada	73.9	69.7	94.9	84.0	81.6	54.1	A
2	Gothenburg	Sweden	72.6	72.9	100.0	85.5	75.9	63.3	B
3	Bergen	Norway	72.1	80.4	100.0	96.8	79.5	48.8	B
4	Raleigh	United States	71.7	77.6	81.6	74.4	63.2	61.5	A
5	Eindhoven	Netherlands	69.6	68.8	98.2	87.3	75.9	42.8	A
6	Ottawa	Canada	69.3	65.6	94.9	84.0	82.9	44.2	A
7	Minneapolis	United States	68.8	75.6	76.4	74.4	55.5	61.5	A
8	Prague	Czechia	68.5	68.9	90.6	77.5	86.1	42.4	A
9	Pittsburgh	United States	68.1	78.6	73.3	74.4	60.4	55.3	A
10	Melbourne	Australia	67.6	49.9	92.7	92.4	69.1	60.5	A
11	Perth	Australia	67.3	56.3	92.7	92.4	69.1	51.2	A
12	Copenhagen	Denmark	67.1	43.7	98.4	81.6	91.0	61.5	A
13	Toronto	Canada	66.8	50.4	94.9	84.0	80.4	54.1	A
14	Graz	Austria	66.8	65.5	90.1	83.1	76.9	41.4	A
15	Brisbane	Australia	66.5	51.8	92.7	92.4	69.1	54.4	A
16	New York	United States	66.4	43.8	85.4	82.0	69.8	75.3	A
17	Adelaide	Australia	66.3	55.5	92.7	92.4	69.1	49.2	A
18	London	United Kingdom	65.6	54.8	82.8	69.4	72.9	59.3	A
19	Braga	Portugal	65.5	64.6	96.7	74.0	68.7	43.8	A
20	Helsinki	Finland	64.9	47.6	93.5	94.5	71.5	52.9	A
21	Jeju City	South Korea	64.9	56.4	98.1	92.3	55.7	50.6	A
22	Chicago	United States	64.8	64.3	66.8	74.4	58.3	61.5	A
23	Incheon	South Korea	64.4	55.9	93.2	92.3	55.7	50.6	A
24	Vancouver	Canada	64.3	45.1	94.9	84.0	78.6	54.1	A
25	Sydney	Australia	64.3	40.7	92.7	92.4	69.1	63.9	A
26	Kaohsiung	Taiwan	64.0	66.8	86.5	90.6	65.4	37.0	A
27	Tallinn	Estonia	63.9	56.4	92.4	67.6	58.4	57.8	A
28	Christchurch	New Zealand	63.9	56.6	94.3	85.7	70.8	41.9	A
29	Wellington	New Zealand	63.8	50.1	94.3	85.7	70.8	48.8	A
30	Lyon	France	63.8	56.3	98.4	71.8	75.1	45.1	A
31	Dunedin	New Zealand	63.7	61.4	94.3	85.7	70.8	37.1	A
32	Porto	Portugal	63.4	58.2	96.7	74.0	68.7	43.8	A
33	Busan	South Korea	63.0	51.9	95.9	92.3	55.7	50.6	A
34	Singapore	Singapore	62.8	44.1	90.4	94.1	45.7	73.3	A
35	Hobart	Australia	62.8	50.2	92.7	92.4	69.1	45.1	A
36	Auckland	New Zealand	62.7	44.4	94.3	85.7	70.8	52.9	A

#	City	Country	SLIC	G	V	Cap	Com	Cr	Grade
37	Suwon	South Korea	62.7	51.4	93.1	92.3	55.8	50.6	A
38	Fukuoka	Japan	62.2	60.0	95.8	74.6	59.9	42.9	A
39	Manama	Bahrain	61.7	72.4	66.9	57.8	42.9	67.5	A
40	Valparaiso	Chile	61.4	63.2	76.4	67.5	66.6	43.0	A
41	Taipei	Taiwan	61.2	45.4	89.5	77.3	65.3	53.7	A
42	Milan	Italy	61.2	77.8	85.8	71.7	66.2	29.4	A
43	Santiago	Chile	61.1	57.4	72.9	77.6	66.6	44.6	A
44	Sapporo	Japan	61.0	62.1	95.8	74.6	59.9	37.9	A
45	Hiroshima	Japan	60.7	61.2	95.8	74.6	59.9	37.9	A
46	Kobe	Japan	60.6	61.0	95.8	74.6	59.9	37.9	A
47	Zurich	Switzerland	60.5	40.4	98.2	78.1	82.1	49.4	A
48	Tokyo	Japan	60.4	49.0	94.2	72.9	73.0	44.2	A
49	Cork	Ireland	59.9	88.1	100.0	86.7	93.4	37.4	C
50	Dubai	United Arab Emirates	59.9	75.7	72.4	76.7	38.5	46.2	A
51	Abu Dhabi	United Arab Emirates	59.8	76.6	70.8	76.7	38.5	46.2	A
52	Amsterdam	Netherlands	59.8	42.9	98.2	87.3	75.9	42.8	A
53	Vienna	Austria	59.5	45.9	90.1	83.1	75.0	41.4	A
54	Venice	Italy	59.2	59.1	84.7	71.4	69.8	34.8	A
55	Bratislava	Slovakia	58.5	60.4	91.2	65.3	64.8	35.3	A
56	Antwerp	Belgium	58.0	73.2	100.0	95.6	88.9	40.4	C
57	Brno	Czechia	57.8	73.5	92.5	100.0	84.6	42.4	C
58	Torun	Poland	57.4	67.2	77.3	77.1	53.5	31.9	A
59	Bologna	Italy	56.9	59.7	87.9	71.7	66.2	29.4	A
60	Katowice	Poland	56.6	65.8	73.7	77.1	53.5	31.9	A
61	Gdansk	Poland	56.5	61.0	83.4	77.1	53.5	31.9	A
62	Port Louis	Mauritius	55.0	59.1	83.6	48.6	50.1	45.4	A
63	Bucharest	Romania	55.0	61.2	85.8	61.0	51.0	34.8	A
64	San Juan	Puerto Rico	54.8	59.3	68.8	68.3	63.6	45.8	B
65	Krakow	Poland	54.2	56.1	74.6	77.1	53.5	31.9	A
66	Bangkok	Thailand	53.7	42.6	76.6	68.2	70.7	38.5	A
67	Ljubljana	Slovenia	53.3	69.3	94.4	100.0	87.7	33.2	C
68	Paris	France	53.2	33.1	98.4	71.8	72.0	45.1	A
69	Montevideo	Uruguay	51.8	53.0	72.0	72.2	71.6	24.4	A
70	Zagreb	Croatia	51.4	67.1	85.0	82.2	73.5	42.3	C
71	San Jose	Costa Rica	51.2	55.3	55.9	56.3	60.9	35.9	A
72	Melaka	Malaysia	50.3	67.0	75.3	49.1	43.9	31.5	A
73	Kota Kinabalu	Malaysia	50.2	65.6	77.9	49.1	43.7	31.5	A
74	Kuching	Malaysia	49.9	65.7	72.0	49.1	44.3	31.5	A

#	City	Country	SLIC	G	V	Cap	Com	Cr	Grade
75	Budapest	Hungary	49.8	48.3	94.7	64.9	60.0	26.2	A
76	Riga	Latvia	49.6	65.4	100.0	66.5	79.2	40.3	C
77	George Town	Malaysia	49.3	62.7	71.1	49.1	44.3	31.5	A
78	Doha	Qatar	49.1	67.2	56.5	58.1	33.4	37.6	A
79	Vilnius	Lithuania	49.1	68.5	100.0	58.4	79.4	41.4	C
80	Male	Maldives	49.1	54.6	80.7	50.8	41.0	35.4	A
81	Panama City	Panama	48.9	71.5	61.2	49.6	51.6	26.9	A
82	Kuala Lumpur	Malaysia	48.1	56.1	72.4	49.1	44.3	31.5	A
83	Jeddah	Saudi Arabia	47.7	95.5	66.7	78.9	24.8	24.1	A
84	Khobar	Saudi Arabia	47.7	94.7	66.7	78.9	24.8	24.1	A
85	Riyadh	Saudi Arabia	47.7	94.5	66.7	78.9	24.8	24.1	A
86	Valencia	Spain	47.4	67.6	100.0	68.9	85.9	30.4	C
87	Curitiba	Brazil	46.5	57.2	58.0	53.9	46.2	27.6	A
88	Hat Yai	Thailand	46.4	53.3	66.7	62.2	49.0	22.8	A
89	Florianopolis	Brazil	46.3	56.2	58.0	53.9	46.2	27.6	A
90	Belgrade	Serbia	46.2	48.9	72.0	59.8	38.0	30.3	A
91	Salalah	Oman	46.1	80.0	71.7	42.0	32.3	30.6	A
92	Cordoba	Argentina	46.1	62.7	78.7	73.1	56.2	10.2	A
93	Sao Paulo	Brazil	45.5	52.3	58.0	53.9	46.2	27.6	A
94	Kuwait City	Kuwait	45.3	93.9	52.9	65.0	41.2	19.9	A
95	Guangzhou	China	45.1	53.2	91.0	74.6	15.8	37.1	A
96	Tianjin	China	45.0	55.7	72.6	74.6	15.8	37.1	A
97	Phuket	Thailand	44.8	46.9	68.0	62.2	49.0	22.8	A
98	Muscat	Oman	44.4	74.8	60.3	42.0	32.3	30.6	A
99	Buenos Aires	Argentina	44.2	55.8	78.7	73.1	56.2	10.2	A
100	Shenzhen	China	43.7	49.2	92.7	74.6	15.8	37.1	A
101	Chiang Mai	Thailand	43.5	51.9	49.7	62.2	49.0	22.8	A
102	Chengdu	China	43.5	48.7	77.0	74.6	16.2	37.1	A
103	Chongqing	China	43.5	49.1	75.2	74.6	16.1	37.1	A
104	Suva	Fiji	42.9	55.3	51.2	57.0	38.2	25.1	A
105	Thimphu	Bhutan	42.7	50.9	46.2	37.3	47.4	34.7	A
106	Hangzhou	China	42.6	45.6	84.0	74.6	16.3	37.1	A
107	Shanghai	China	42.3	45.3	85.2	74.6	15.8	37.1	A
108	Rabat	Morocco	42.2	51.8	66.7	34.1	27.6	41.2	A
109	Casablanca	Morocco	41.8	51.4	62.3	34.1	27.6	41.2	A
110	Tbilisi	Georgia	41.0	55.7	77.6	66.9	47.0	43.1	C
111	Nanjing	China	40.3	66.0	75.3	58.9	5.2	34.5	A
112	Aqaba	Jordan	39.7	48.5	62.5	46.5	14.3	40.8	A
113	Arequipa	Peru	39.1	62.7	40.8	38.5	43.0	24.7	A
114	Amman	Jordan	39.0	46.7	59.0	46.5	14.3	40.8	A

#	City	Country	SLIC	G	V	Cap	Com	Cr	Grade
115	Bogota	Colombia	38.9	55.2	39.2	50.7	48.6	19.6	A
116	Medellin	Colombia	38.1	58.3	35.2	50.7	48.6	19.6	A
117	Lima	Peru	38.0	59.5	37.3	38.5	43.0	24.7	A
118	Nizhny Novgorod	Russia	36.3	67.0	70.8	65.6	13.9	15.5	A
119	Bengaluru	India	36.2	46.8	54.5	46.0	40.0	14.6	A
120	Hyderabad	India	36.0	47.0	52.7	46.0	40.0	14.6	A
121	Makati	Philippines	35.8	56.9	45.7	38.9	36.8	17.5	A
122	Cebu City	Philippines	35.8	59.5	44.8	38.9	36.8	17.5	A
123	Pune	India	34.8	46.5	44.8	46.0	40.0	14.6	A
124	Chandigarh	India	34.7	46.0	44.1	46.0	40.3	14.6	A
125	Munich	Germany	34.4	30.3	100.0	72.9	78.7	36.2	C
126	Surabaya	Indonesia	33.8	67.0	36.2	39.1	36.4	17.8	A
127	Da Nang	Vietnam	33.3	58.3	77.6	49.6	22.3	20.8	B
128	Jakarta	Indonesia	33.3	63.7	34.5	39.1	36.4	17.8	A
129	Gaborone	Botswana	33.2	52.4	45.3	6.7	52.3	39.0	A
130	Windhoek	Namibia	33.1	44.0	51.9	16.5	50.5	25.4	A
131	Merida	Mexico	33.0	60.1	29.3	48.7	46.3	14.1	A
132	Santo Domingo	Dominican Republic	32.6	55.2	88.8	46.3	51.8	28.2	C
133	Cape Town	South Africa	32.3	31.8	35.1	26.3	53.6	29.8	A
134	Port Vila	Vanuatu	32.0	43.3	100.0	73.3	60.6	21.9	C
135	Johannesburg	South Africa	31.7	33.0	31.1	26.3	53.6	29.8	A
136	Mexico City	Mexico	31.7	52.3	27.1	48.7	46.3	14.1	A
137	Guadalajara	Mexico	31.5	57.8	25.8	48.7	46.3	14.1	A
138	Moscow	Russia	31.4	44.8	71.2	65.6	13.9	15.5	A
139	Kandy	Sri Lanka	31.4	48.3	49.4	34.2	35.4	11.7	A
140	Colombo	Sri Lanka	31.0	45.8	48.5	34.2	35.4	11.7	A
141	Phnom Penh	Cambodia	29.6	54.6	55.1	18.2	25.5	36.3	B
142	Kigali	Rwanda	29.2	59.5	47.3	14.2	27.7	24.8	A
143	Pokhara	Nepal	28.1	48.9	36.5	22.4	31.9	16.6	A
144	Kathmandu	Nepal	26.7	47.8	30.3	22.4	31.9	16.6	A
145	Nairobi	Kenya	26.4	54.2	53.9	13.0	34.9	14.5	A
146	Dhaka	Bangladesh	25.3	51.4	36.9	19.8	24.1	15.7	A
147	Chattogram	Bangladesh	25.2	52.9	36.9	19.8	24.1	15.7	A
148	Accra	Ghana	21.8	59.0	29.0	8.1	41.0	17.8	A
149	Kumasi	Ghana	21.1	60.7	27.5	8.1	41.0	17.8	A
150	Dakar	Senegal	20.8	44.4	12.6	13.0	37.4	26.7	A
151	Asuncion	Paraguay	17.5	52.9	96.3	25.3	49.3	14.9	C
152	Dar es Salaam	Tanzania	17.3	42.4	70.1	40.2	34.2	12.7	C
153	Islamabad	Pakistan	15.4	46.3	25.0	18.8	18.7	4.4	A
154	Karachi	Pakistan	15.4	46.5	25.0	18.8	18.7	4.4	A

#	City	Country	SLIC	G	V	Cap	Com	Cr	Grade
155	Lahore	Pakistan	15.4	46.8	25.0	18.8	18.7	4.4	A
156	Cuenca	Ecuador	11.9	51.5	100.0	21.2	55.2	8.0	C
157	Kampala	Uganda	0.1	42.1	0.0	31.3	23.4	22.5	C
158	Alexandria	Egypt	0.0	49.5	0.0	88.7	10.8	30.6	C